

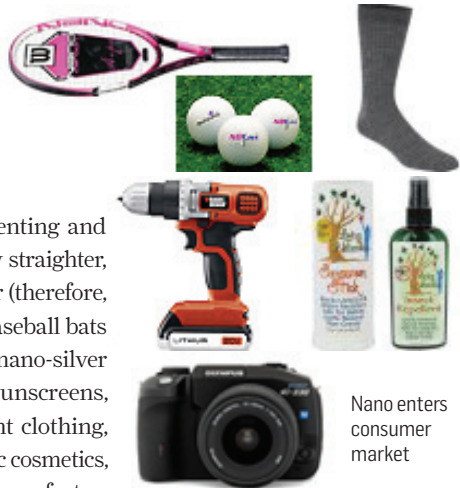
and led directly to formation of the U.S. National Nanotechnology Initiative in 2000.

Dip-Pen Nanolithography (1999)

A pivotal development in the constellation of nanotechnology tools was Dip-Pen Nanolithography or DPN. Invented in 1999 by Chad A. Mirkin (Rathmann Professor of Chemistry and Director of the Institute for Nanotechnology at Northwestern University), the concept is based upon a classic quill pen a 4,000 year old technology. Using an atomic microscope tip, DPN allows researchers to precisely lay down or “write” chemicals, metals, biological macromolecules, and other molecular “inks” with nano dimensions and precision on a surface. DPN has opened the door to credible nano-manufacturing techniques for smaller, lighter weight, faster, and more reliably produced electronic circuits and devices, high-density storage materials, and biological and chemical sensors.

Consumer Products (1999-2000)

By the turn of the twenty-first century, Consumer products making use of nanotechnology began appearing in the marketplace, including lightweight nanotechnology-enabled automobile bumpers that resist denting and scratching, golf balls that fly straighter, tennis rackets that are stiffer (therefore, the ball rebounds faster), baseball bats with better flex and “kick,” nano-silver antibacterial socks, clear sunscreens, wrinkle- and stain-resistant clothing, deep-penetrating therapeutic cosmetics, scratch-resistant glass coatings, faster-recharging batteries for cordless electric tools, and improved displays for televisions, cell phones, and digital cameras.



Nano enters consumer market

National Nanotech Initiative (2000)

In August 1999, the Interagency Working Group on Nanotechnology completed its first draft of a plan for an initiative in defining the art of nanoscale